

ThoughtWorks®

June 2014 - v0.2

Cross Functional Requirements

...or ~~Non Functional Requirements~~

READ ME FIRST: HOW TO USE THIS DECK

Practical things first:

- *You can print two-sided cards with questions or decide to print only the even pages*
- *You can facilitate exercises in various formats to prioritize and/or discuss the most relevant*
- *You can use the questions on the back of the card to start a conversations*
The existing questions are only a first draft, don't expect them to be comprehensive
- *Add your own cards or your own questions*

Understand how to use the cards reading some articles on myThoughtWorks:

- The original Cross Functional Requirement method cards and facilitation exercises (read the comments): <https://my.thoughtworks.com/docs/DOC-977>
- Non Functional Requirements: <https://my.thoughtworks.com/docs/DOC-14605>
- Non-functional Requirements - Guidance & Sample Questions: <https://my.thoughtworks.com/docs/DOC-1971>
- AAA Toolbox: Cross-functional Requirements: <https://my.thoughtworks.com/blogs/lous/2012/05/14/aaa-toolbox-cross-functional-requirements>
- Type of Non-Functional Stories.xls: <https://my.thoughtworks.com/docs/DOC-11886>
- Understanding Non Functional Requirements: <https://my.thoughtworks.com/docs/DOC-2506>

Accessibility

To all your users

- *Do we need to support users with special needs and disabilities?*
- *Do we need to support any accessibility standard? WAI-WCAG, OAF, etc.?*
- *What “all” users mean? Mobile, screen readers, web, desktop?*
- *?*

Archiving

How long to store what and why?

How to store?

- *How are backups performed ?*
- *Who, When, How access to historical information ?*
- *How historical information are recovered ?*
- *?*

Auditability

Who did what and when

- *Who has access to the audit information?*
- *What operations should be tracked?*
- *Are there any regulation or legal requirement?*
- *Sometimes auditability disguise specific user feature...*
- *?*

Authentication

How do we ensure “you are who you say you are”

- *How are user identified?*
- *Are existing authentication system present?*
- *Single Sign On standards?*
- *?*

Authorization

How do we ensure “which user access to what”

- *Do we need to restrict available features, resources, pages?*
- *Do we need to maintain groups and roles?*
- *Who maintains this access information? And where? LDAP, AD, custom?*
- *To which level of depth and granularity we limit access? Features, web pages, APIs, database fields*
- *?*

Availability

When and for who?

- *Is there a target defined 99.9**% ?*
- *Is the system used 24/7 ? Or in which period of the day/week is not/less used?*
- *What is the impact of scheduled downtimes?*
- *Or specific critical deadlines during the year/month?
eg. Black Friday?*
- *What HA infrastructure is already in place?*
- *Remember the CAP theorem: Consistency, Availability, Partition tolerance*

Compatibility

Adherence to industry or existing standard for input/outputs formats and protocols

- *With what system we need to interoperate with?*
- *Is in this industry is use a specific standard?*
- *Is existing data stored in specific formats?*
- *?*

Configurability

Ability for end users or administrator to change aspects of the software configuration easily

- *Note: to not be confused with 'feature toggles' (that are only used as a development practice)*
- *Can users or administrator configure features behaviors?*
- *Do we need a central configuration or feature specific? And how is it managed?*
- *?*

Continuity

Ability for business processes to continue in the event of system downtime

- *Disaster recovery plans?*
- *Support for client side decentralised operations?*
- *?*

Data integrity & consistency

- *Tolerance for loss and corruption of data?*
- *Checksum, journaling, data recovery mechanism?*
- *Consistency across nodes (CAP theorem)*
- *?*

Distributability

*Ability to distribute components onto different tiers
or nodes*

- *How many tiers levels are necessary or we could leverage?*
- *What is the level of flexibility required? And who might be other systems using lower tiers?*
- *Are necessary separate CFR requirements for each layer?*
- *?*

Extensibility

How important might be to plug new pieces of functionality

- *Do we need to allow other groups to develop plugins?
Or only the existing team extends new functionalities?*
- *Can the end users program the system itself?*
- *?*

Help & support

- *Do we need user documentation?*
- *Do we need help features as part of the system?*
- *Do we need to involve a help and support team?*
- *Support documentation or training?*
- *?*

Installability & deployment

- *What is the required ease of system installation?*
- *What are the supported platforms (OS, VMs versions)?*
- *To which extent we need to push deploy automation? one-click deploy? Continuous deployment?*
- *How do we rollback?*
- *?*

Integratability & Interoperability

- *Ability for the application/solution to easily fit as part of a larger system.*
- *Ease of use of the APIs (not about standards and protocols)?*
- *?*

Legal compliance

- *What legislative constraints is the system operating in?*
- *Data protection standards?*
- *Regulations regarding how the application is developed or deployed?*
- *User policies and regulations that limit the features been developed?*

Leverageability & reuse

- *Creation of enterprise components to reuse across multiple products?*
- *Can we reuse existing enterprise platform components?*
- *Building stepping stones to use in different phases?*
- *?*

Localization & Internationalization

Support for multiple languages

- *Multilanguage support for static system text?*
- *Multilanguage support for operational data?*
- *Date, time, currency internationalization?*
- *Conversions and translations?*
- *Do we need external translators?*
- *?*

Monitoring

- *Are there existing tools and infrastructure in place?*
- *Should we use monitoring tools for business metrics?*
- *Visibility and who has access to monitoring informations?*
- *What types of alert and notifications should be based on monitoring data?*
- *?*

Multiple environments support

- *Development, Testing, Staging, Production?*
- *What type of infrastructure is necessary for each environment?*
- *?*

Performance

- *Throughput and response times?*
- *Peak number of users, peak time of the year, ...?*
- *Maximum load before the system start failing?*
- *Needs for parallel or asynchronous processing?*
- *When we prefer realtime vs scheduled batches operations?*
- *Improve performance incrementally or hard requirement?*

Portability

- *How easy should be to move the application to a different database or platform?*
- ?

Data privacy

- *What data are critical and should not be visible to internal operators and other users?*
- *Should some data be encrypted?*
- *Can Dev / Testing environment have copies of production data?*
- *Do we need anonymization tools?*
- *Can we minimize the amount of time we keep history of personal private data?*
- *?*

Resilience & fault tolerance

- *In which ways the can system downgrade functionalities if a component is unavailable?*
- *How quickly can the system recover in case of disaster? Backup strategy?*
- *Chaos monkey?*
- *?*

Reliability

- *How confident we need to be about the accuracy of the system? Space shuttle or web game app?*
- *What is the business cost of an inaccuracy?*
- *?*

Scalability

- *Horizontal*
- *Do we need to increase/decrease the allowed throughput based on changing volume of data or users?*
- *?*

Partition tolerance

- *Remember the CAP theorem: Consistency, Availability, Partition tolerance*
- *Should the entire set of data be available during downtime of a partition? (Resilience)*
- *Can a partition recover from a loss of synchronization? eg. Network unavailable (Resilience)*
- ?

Security

- *What security audit strategies we need to put in place?*
- *Do we need security auditing? Who will conduct penetration testing?*
- *What are there enterprise guidelines?*
- *Should the traffic or data be encrypted?*
- *What are the valuable information we need to protect?*
- *?*

Supportability

- *What levels of support (customer/tech) are in place and what information do they have access to?*
- *Do we need training and documentation for tech support?*
- *Are additional type of logging needed?*
- *Any low level operations or access available to support?*
- *?*

Usability & User Experience

- *How important is usability and UX for the business?*
- *How quickly we need to iterate on UX changes ?*
- *Are there company UX guidelines to follow?*
- *How many dedicated skills we need?*
- *?*

Versioning & upgradeability

- *Do we need major, minor, build versions and what they communicate?*
- *How should we track dependencies across internal components/systems?*
- *And external ones?*
- *What type of backward compatibility constraints?*
- *?*

THANKS

Modify and reuse freely (within ThoughtWorks)

*For questions, suggestions and feedback:
...link...*

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